NODE 1 AND NODE 3 INTEGRATION — STRUCTURAL CONTEXT AND SIZE-STATE INTERPRETATION

System-Level Structural Context Concept

The longitudinal penile physiology monitoring device (Node 1) may operate in conjunction with a penile geometry or footprint scanning device (Node 3) configured to establish a structural reference model representing anatomical dimensions, proportional geometry, and baseline spatial characteristics. Data generated by Node 3 may provide contextual calibration allowing Node 1 measurements to be interpreted relative to individualized anatomical structure rather than generalized assumptions.

Size-State Determination Framework

Node 1 may determine the physiologic size state of the monitored anatomy by referencing geometric profiles obtained from Node 3. Structural parameters including baseline length, circumference distribution, proportional zones, symmetry characteristics, or deformation response profiles may enable classification of physiologic states such as flaccid baseline, transitional expansion, functional erection, partial rigidity, or maximum activation relative to individualized structural capacity.

Rather than interpreting deformation alone, Node 1 may evaluate percent-of-capacity expansion, structural engagement ratios, and relative dimensional activation derived from Node 3 reference geometry.

Calibration and Normalization

Structural data obtained from Node 3 may support normalization of rigidity and perfusion metrics across users with differing anatomical dimensions. Calibration procedures may adjust sensor interpretation thresholds, deformation sensitivity, and rigidity classification boundaries according to individualized geometry models.

Dynamic Structural Modeling

Integration between Node 1 and Node 3 may enable dynamic modeling in which real-time physiologic measurements are mapped onto previously captured geometric representations. Computational systems may generate estimates of volumetric change, structural engagement progression, or proportional activation patterns during physiologic events.

Extended Applications

Combined Node 1 and Node 3 datasets may support device fitting optimization, personalized accessory sizing, longitudinal structural change monitoring, compatibility analysis with partner-associated systems, and advanced physiologic analytics derived from interaction between structure and function.

The integration embodiments described herein are illustrative and non-limiting and encompass any computational framework linking structural geometry characterization with longitudinal physiologic monitoring.